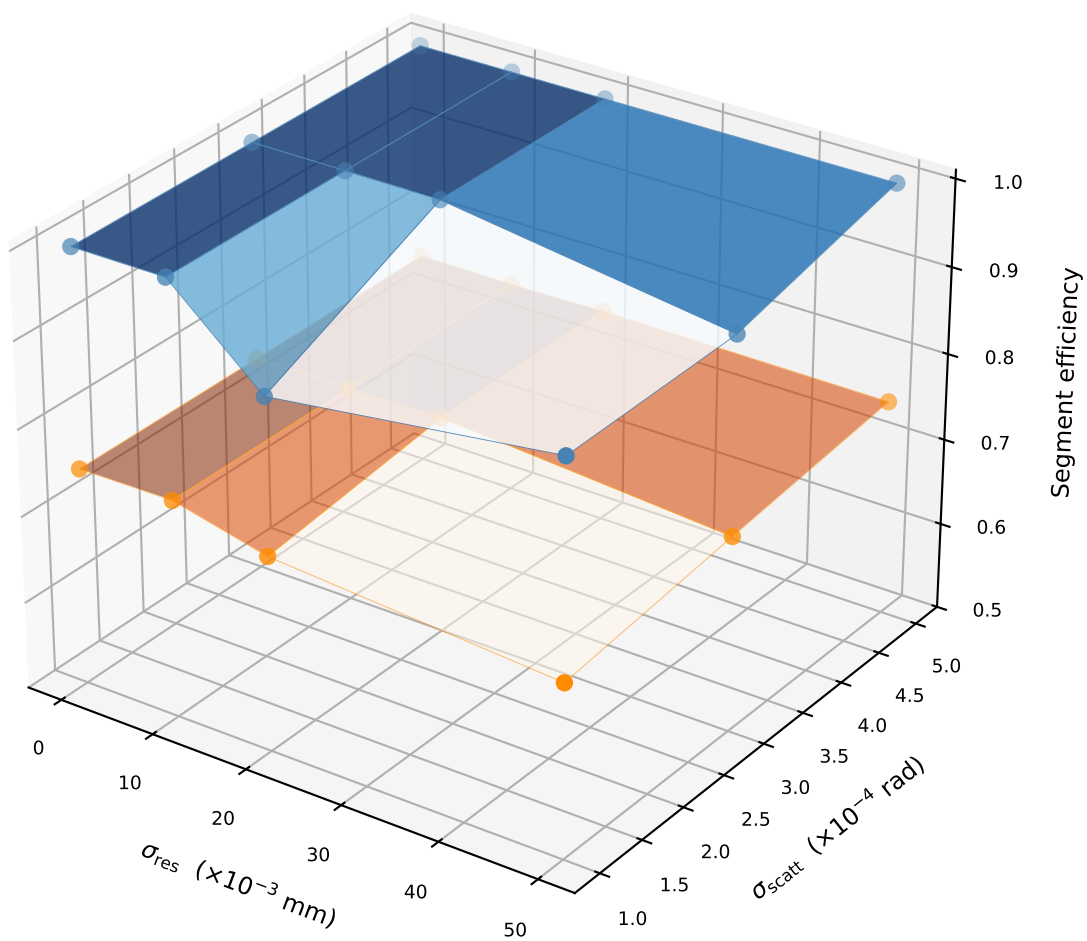
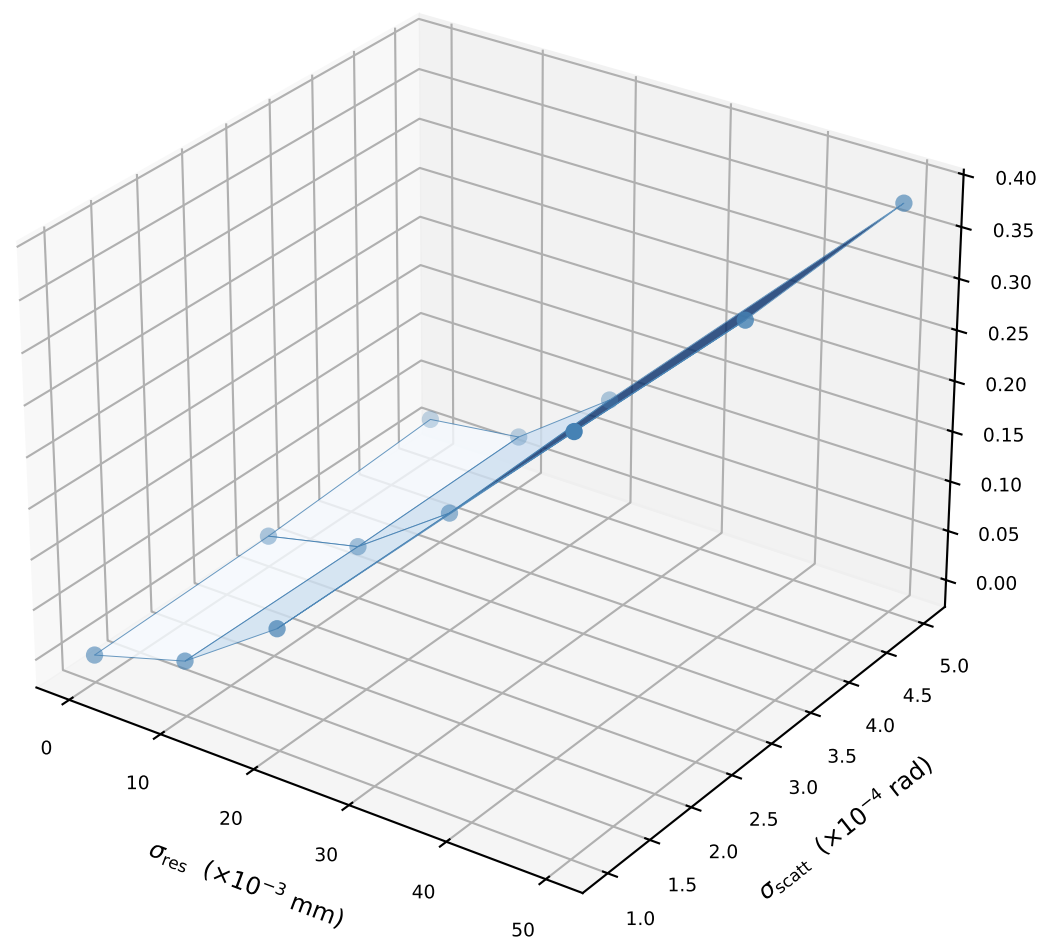


# Metrics over ( $\sigma_{\text{res}}, \sigma_{\text{scatt}}$ ) phase space — $T = 50$ tracks (reps/cell: 10-10)

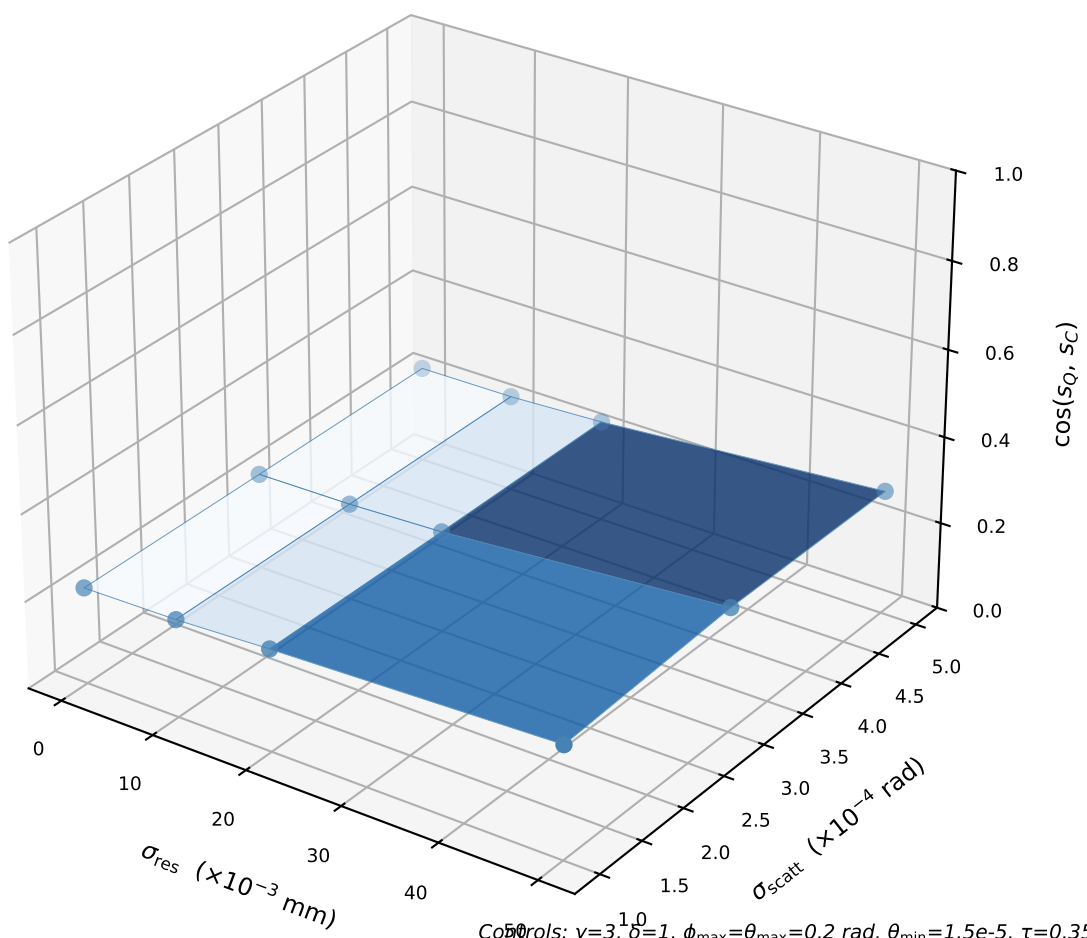
Segment efficiency (Blue=C, Orange=Q)



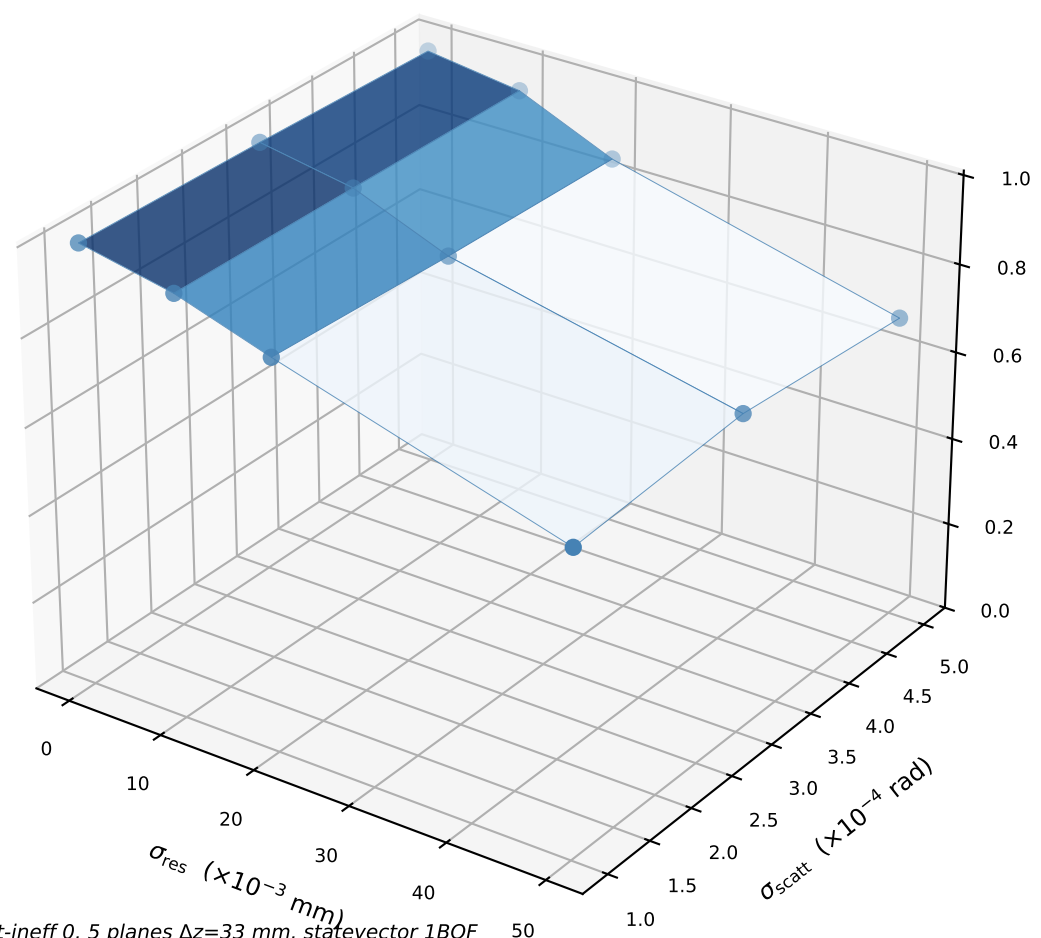
Angle false rate (Classical)



$\cos(s_Q, s_C)$  (Classical)



Quantum purity (Classical)



Controls:  $\gamma=3, \delta=1, \phi_{\text{max}}=\theta_{\text{max}}=0.2$  rad,  $\theta_{\text{min}}=1.5e-5, \tau=0.35, \epsilon=\text{calc.}(s=3), \text{conv OFF, hit-ineff } 0, 5 \text{ planes } \Delta z=33 \text{ mm, statevector 1BQF}$   
 Fixed here:  $T=50$  fixed; Blue=Classical, Orange=Quantum  
 Swept:  $\sigma_{\text{res}} \times \sigma_{\text{scatt}}$  (full grid)